

Keith's Mock Exam for MA1200 Set 1

This document is primarily intended for the uses of students registered for PALSI at CityUHK, and participated in Keith TSE's section.

Last updated in November 2023.

Rules

Time allowed: THREE hours.

Attempt ALL questions.

Total marks: 100.

Question 1

Differentiate with respect to x :

(a) $\frac{(3x+2)^5}{x^4}$ (2 marks)

(b) $e^{\sin 4x}$ (2 marks)

(c) $\sinh x^4$ (3 marks)

(d) $\tan^{-1}(4x^2 - x)$ (3 marks)

(e) $\log_e(\log_e \sin x)$ (3 marks) (Hint: $-\frac{\tan x}{\log_e \cos x}$ is a wrong answer for Question 1(e).)

Question 2

(a) Let $f(x) = |\cot x|$ for $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Is $f(x)$ differentiable at $x = 0$? Give your reason.

(Hint: $|\cot x| = \begin{cases} \cot x, & \text{if } x \in [0, \frac{\pi}{2}) \\ -\cot x, & \text{if } x \in (-\frac{\pi}{2}, 0) \end{cases}$) (4 marks)

(b) A curve has parametric equations $x = \cos t + \log_e \tan(\frac{t}{2})$, $y = \sin t$, where t is the parameter and $0 < t < \pi$.

Show that

i) $\frac{dy}{dx} = \tan t$,

ii) $\frac{d^2y}{dx^2} = \frac{\sin t}{\cos^4 t}$. (5 marks)

Question 3

(a) Evaluate the following limits:

(i) $\lim_{x \rightarrow 0} \frac{2-2x^3-2\cos x}{x^5},$

(ii) $\lim_{x \rightarrow 1} (x-1) \cot\left(\frac{\pi x}{2}\right).$ (8 marks)

(b) Show from first principles that $\frac{d}{dx}(\cos x) = -\sin x.$

(Hint: $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}.$) (5 marks)

Question 4

Given a unit circle, we try to calculate the perimeters of inscribed regular 5-sided, 10-sided, 20-sided, ..., polygons, thus obtaining lower bounds for the perimeter of the unit circle ($= 2\pi$ units).

Let P_n denote the perimeter of inscribed regular n -sided polygon.

(a) Sketching, or otherwise, describes the geometrical setup. Find the values of P_5 , P_{10} , P_{20} and P_{40} , correct your answers to 3 decimal places. (10 marks)

(b) Find the value of $\frac{64P_{40}-32P_{20}+4P_{10}-P_5}{42}$, correct your answer to 2 decimal places. (3 marks)

(Hint: The result in Question 4(b) is an approximation to the value of π .)

Question 5

(a) If $y = (bx + c)^q$, where a and q are positive integers, b is a constant, find the general formula for the n th derivative of y with respect to x . (4 marks)

(b) Express $\frac{5x^2}{(2x-1)^3}$ in partial fractions. (5 marks)

(c) Hence, or otherwise, find $\frac{d^t}{dx^t} \left(\frac{5x^2}{(2x-1)^3} \right)$. You need not simplify your answer. (4 marks)

Question 6

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $f(x)$ is defined as $f(x) = (x^2)^{1/3} - (x^2 - 9)^{1/3}$ for $x \in \mathbb{R}$.

(a) Show that $f(x)$ is an even function of x and that $f(x) > 0$ for $x \in \mathbb{R}^+$. (4 marks)

(b) Expand and simplify $[x^3 - (x^2 - 3)^{1/3}] [x^3 + (x^2 - 3)^{1/3}]$. (2 marks)

(c) Evaluate $\lim_{x \rightarrow +\infty} f(x)$. (3 marks)

(d) Solve the equation $f'(x) = 0$. (4 marks)

Question 7

Let $\frac{x^2}{9} - y^2 = 1$ be the equation of the hyperbola H , and let $\left(t + t^{-1}, \frac{t-t^{-1}}{2}\right)$ be the coordinates of a point P , where $t \neq 0$.

- (a) Find the coordinates of the foci and vertices of the hyperbola H . (5 marks)
- (b) Show that the point P lies on the hyperbola H . (3 marks)
- (c) Find the equation of the tangent to the hyperbola H at point P . (5 marks)

Question 8

Let $y = \cos(3 \log_e(1+x))$, for $x > -1$.

- (a) Show that $(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + 9y = 0$. (3 marks)
- (b) Hence, deduce that, $(1+x)^{n+2} \frac{d^{n+2} y}{dx^{n+2}} + (2n+1)(1+x)^{n+1} \frac{d^{n+1} y}{dx^{n+1}} + (n^2+9)(1+x)^n \frac{d^n y}{dx^n} = 0$, for $n = 1, 2, \dots$ (5 marks)
- (c) Hence, or otherwise, find the power series for $\cos(3 \log_e(1+x))$ in ascending powers of x , up to and including the term in x^4 . (5 marks)

(Hint: Leibniz' rule: For any functions u and v whose derivatives up to the n th order exist,

$$(uv)^{(n)} = C_n^0 u^{(n)} v + C_n^1 u^{(n-1)} v^{(1)} + \dots + C_n^n u v^{(n)},$$

where $C_n^r = \frac{n!}{(n-r)!r!}$, $u^{(r)}$ and $v^{(r)}$ are the r th derivatives of u and v respectively, for $r = 1, 2, \dots, n$.)